

Teaching Exam

Subject – Computer Science

Topic – Digital Logic



Lecture No.- 3

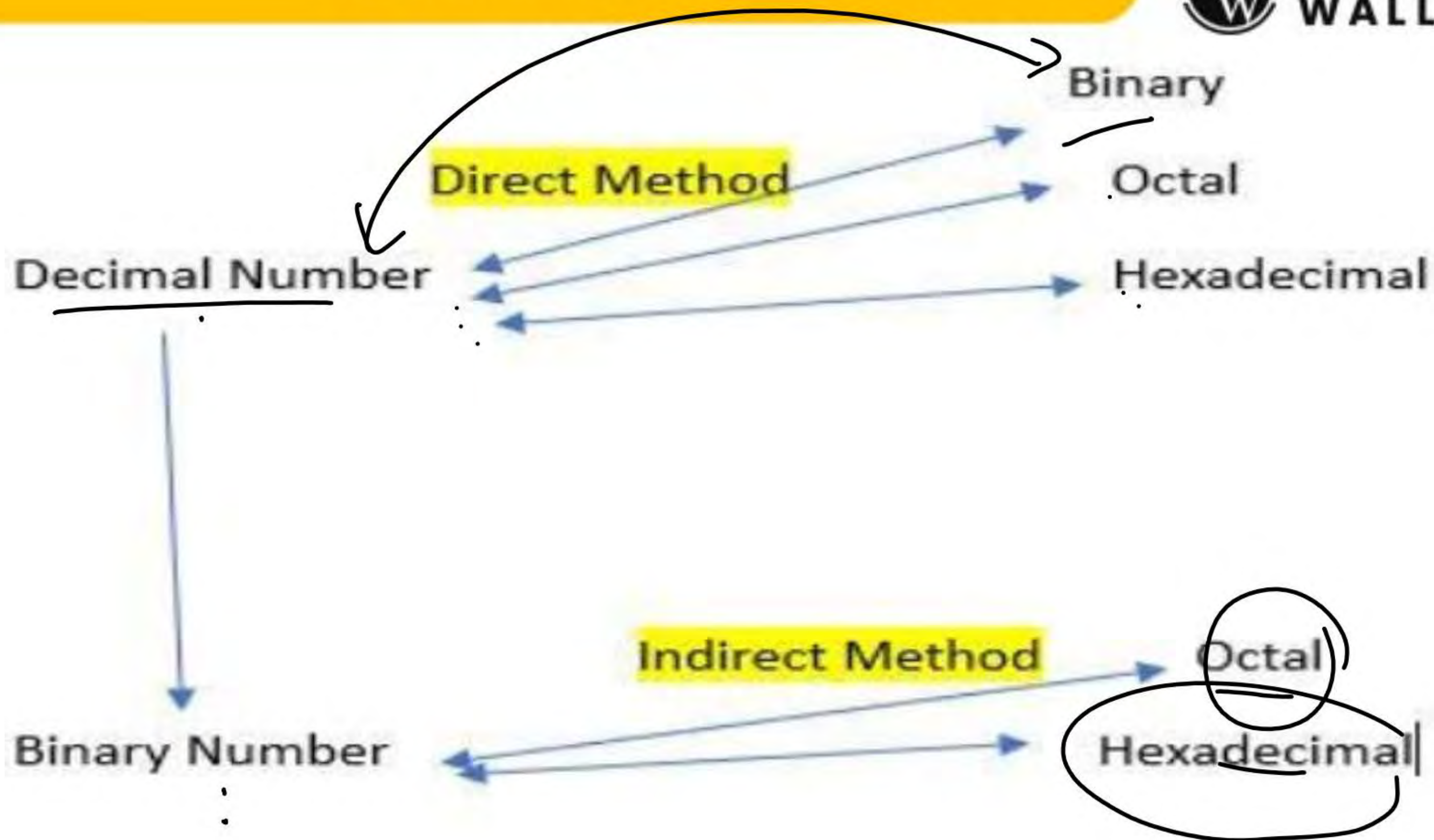
By- Rahul Sir

Topics to be Covered

Topic



- 1:- Decimal To Binary Conversion
- 2:- Binary To decimal Conversion
- 3:- Decimal To octal Conversion
- 4:- Octal to decimal Conversion
- 5:- Decimal To Hexadecimal Conversion
- 6:- Hexadecimal to Decimal Conversion
- 7:- Example on convert Decimal point value to binary, octal , hexadecimal ✓
- 8:- Binary to octal Conversion ✓
- 9:- Binary to Hexadecimal Conversion ✓
- 10:- Binary Addition ✓
- 11:- Binary Subtraction ✓
- 12:- Binary subtraction using 1's complement ✓
- 13:- Binary Subtraction using 2's complement ✓





How to use Binary to Decimal Calculator?

We can easily use the **binary-to-decimal calculator** by following the steps discussed below,

Step 1: Enter the given value in the binary input field.

Step 2: Click on the convert button to convert the binary value into the decimal value.

Step 3: The value shown as the result is the required value in the decimal form.

Binary to Decimal Formula

To convert a **binary number to decimal** we need to perform a multiplication operation on each digit of a binary number from right to left with powers of 2 starting from 0 and add each result to get the **decimal** number of it.

Decimal Number = n^{th} bit $\times 2^{n-1}$

Binary to Decimal Formula

$$n = b_nq + b_{n-1}q^{n-2} + \dots + b_2q^2 + b_1q^1 + b_0q^0 + b_{-1}q^{-1} + b_{-2}q^{-2}$$



Convert Binary To Decimal Number System :-

1:- $(10111000)_2 = (?)_{10}$

2:- $(10101010)_2 = (?)_{10}$

3:- $(0000100001)_2 = (?)_{10}$



How to Convert Decimal to Binary

To convert decimal numbers into binary, there are several methods, including formulas and division techniques. In this explanation, we'll use the remainder method. The steps to convert a decimal number to binary using this method are as follows:

Step 1: Divide the given decimal number by 2, and find the remainder (R_i).

Step 2: Now divide the quotient (Q_i) that is obtained in the above step by 2, and find the remainder.

Step 3: Repeat the above steps 1 and 2 until 0 is obtained as a quotient.

Step 4: Write down the remainder in the following manner: the last remainder is written first, followed by the rest in reverse order ($R_n, R_{(n-1)}, \dots, R_1$). Thus binary conversion of the given decimal number will be obtained.



Convert Decimal to Binary Number System:-

1:- $(789)_{10} = (?)_2$

2:- $(1024)_{10} = (?)_2$

3:- $(9999)_{10} = (?)_2$



Convert Decimal to Octal with Steps

Follow the steps given below to learn the decimal to octal conversion:

1. Write the given decimal number
2. If the given decimal number is less than 8 the octal number is the same.
3. If the decimal number is greater than 7 then divide the number by 8.
4. Note the remainder, we get after division
5. Repeat step 3 and 4 with the quotient till it is less than 8
6. Now, write the remainders in reverse order (bottom to top)
7. The resultant is the equivalent octal number to the given decimal number.

$$\begin{array}{r} 8 \overline{) 1792} \\ \underline{16} \\ 19 \\ \underline{16} \\ 32 \\ \underline{32} \\ 0 \end{array}$$

For example: Convert 1792 into an octal number.

Decimal Number	Operation	Quotient	Remainder	Octal Number
1792	$\div 8$	224	0	0
224	$\div 8$	28	0	00
28	$\div 8$	3	4	400
3	$\div 8$	0	3	3400



1:- Convert $(127)_{10}$ to Octal.

2:- Convert $(52)_{10}$ to octal.

3:- Convert $(456)_{10}$ to Octal

$$\begin{array}{r} 8 \overline{) 127} \\ \end{array}$$



Octal to Decimal

To convert an octal number to a decimal number we need to multiply each digit of the given octal with the reducing power of 8.

Let us learn here, the conversion of Octal number to Decimal Number or base 8 to base 10.

Examples on Octal to Decimal

Example 1: Suppose 215_8 is an octal number, then its decimal form will be,

$$\begin{aligned} 215_8 &= 2 \times 8^2 + 1 \times 8^1 + 5 \times 8^0 \\ &= 2 \times 64 + 1 \times 8 + 5 \times 1 = 128 + 8 + 5 \\ &= 141_{10} \end{aligned}$$

Example 2: Let 125 is an octal number denoted by 125_8 . Find the decimal number.

$$\begin{aligned} 125_8 &= 1 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 \\ &= 1 \times 64 + 2 \times 8 + 5 \times 1 = 64 + 16 + 5 \\ &= 85_{10} \end{aligned}$$



Decimal Number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Equivalent Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Decimal to Hexadecimal Conversion With Steps



Go through the steps given below to learn how to convert the numbers from decimal to hex.

Step 1: First, divide the decimal number by 16, considering the number as an integer.

Step 2: Keep aside the remainder.

Step 3: Again divide the quotient by 16 and repeat till you get the quotient value equal to zero.

Step 4: Now take the values of the remainder's left in the reverse order to get the hexadecimal numbers.

Note: Remember, from 0 to 9, the numbers will be counted as the same in the decimal system. But from 10 to 15, they are expressed in alphabetical order such as A, B, C, D, E, F and so on.



Topic : Decimal to HexaDecimal Number

A → 10
B → 11
C → 12
⋮
F → 15

10 → A
11 → B
12 → C
13 → D
14 → E
15 → F

(312)₁₆

1:- Convert (960)₁₀ into hexadecimal.

(312)₁₆

2:- Convert (786)₁₀ into hexadecimal.

(650)₁₆

3:- Convert (1616)₁₀ into hexadecimal.

16 | 1616
16 | 101
 0
 (650)₁₆

16 | 786
16 | 49
 3
 (312)₁₆

(312)₁₆

16 | 960
16 | 60
 0
 (784)₁₆

16 | 150
16 | 9
 6
 (866)₁₆

16 | 960
16 | 60
 0
 (3120)₁₆

11 → D
12 → B

752
16 | 752
16 | 47
 8
 (4A0)₁₆

16 | 40
16 | 2
 0
 (28)₁₆



Topic : ~~Decimal to HexaDecimal Number~~



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~~Hexadecimal to Decimal~~

As we know, number systems can be converted from one base to another. Thus, we can convert hexadecimal numbers to decimal easily. This **number system conversion** can be done as explained in the example given below:

Example:

Convert 7CF (hex) to decimal.

Solution:

Given hexadecimal number is 7CF.

In hexadecimal system,

$$7 = 7$$

$$C = 12$$

$$F = 15$$

$$\begin{array}{r} 3 \quad 4 \quad 6 \\ 256 \\ \times 7 \\ \hline 1792 \end{array}$$

$$\begin{array}{ccc} 16^2 & 16^1 & 16^0 \\ 7 & C & F \\ \downarrow & \downarrow & \downarrow \\ 7 & 12 & 15 \end{array} \Rightarrow 7 \times 16^2 + 12 \times 16^1 + 15 \times 16^0$$
$$\Rightarrow 1792 + 192 + 15$$
$$\Rightarrow (1999)_{10}$$

To convert this into a decimal number system, multiply each digit with the powers of 16 starting from units place of the number.

$$7CF = (7 \times 16^2) + (12 \times 16^1) + (15 \times 16^0)$$

$$= (7 \times 256) + (12 \times 16) + (15 \times 1)$$

$$= 1792 + 192 + 15$$

$$= 1999$$



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$$= (7 \times 256) + (12 \times 16) + (15 \times 1)$$

$$= 1792 + 192 + 15$$

$$= 1999$$



$(7086)_{10}$

$$\begin{array}{r} 1^2 \\ 4096 \\ 3328 \\ \hline 166 \end{array}$$

$$\begin{array}{r} 16^3 \\ 2 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 16^2 \\ D \\ \hline 13 \end{array}$$

$$\begin{array}{r} 16^1 \\ A \\ \hline 10 \end{array}$$

$$\begin{array}{r} 16^0 \\ 6 \\ \hline 6 \end{array}$$

$$256 \times 13$$

$$1 \times 16^3 + 13 \times 16^2 + 10 \times 16^1 + 6 \times 16^0$$

$$\Rightarrow 4096 + 3328 + 160 + 6 = 7590$$

1:- Convert $(1DA6)_{16}$ to decimal system

2:- Convert $(E8B)_{16}$ to decimal system

3:- Convert $(FFF)_{16}$ to decimal system

$$\begin{array}{r} 1^2 \\ 2056 \\ 768 \\ \hline 2824 \end{array}$$

$$\begin{array}{r} 1 \\ 2056 \\ 48 \\ \hline 2104 \end{array}$$

$$\begin{array}{r} 3 \\ 256 \\ \times 16 \\ \hline 1536 \\ 256 \times \\ \hline 4096 \end{array}$$

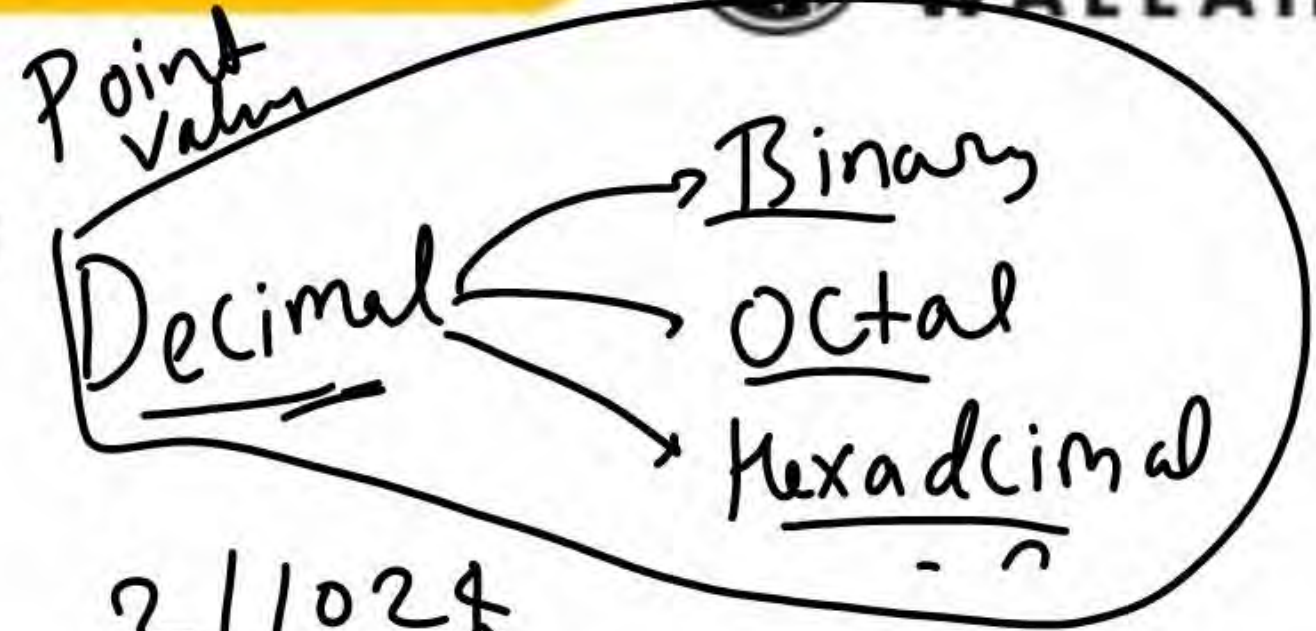
$$\begin{array}{r} 1 \\ 256 \\ \times 3 \\ \hline 768 \end{array}$$



$$\begin{array}{r} 0.690 \\ \times 2 \\ \hline 1.380 \\ \times 2 \\ \hline 2.760 \\ \times 2 \\ \hline 5.520 \\ \times 2 \\ \hline 11.040 \\ \times 2 \\ \hline 22.080 \end{array}$$

0.75 $\times 2 = 1.5$

$$\begin{array}{r} 0.1024 \\ \times 2 \\ \hline 0.2048 \end{array} \quad \text{K.W}$$



1:- Convert $(102.25)_{10}$ to Binary system

2:- Convert $(786.345)_{10}$ to Binary system

3:- Convert $(1024.1024)_{10}$ to Octal system

$$\begin{array}{l} 0.1024 \times 2 = 0.2048 \rightarrow 100000000000.0000 \\ 0.2048 \times 2 = 0.4096 \\ 0.4096 \times 2 = 0.8192 \rightarrow 0 \end{array}$$

2	1024	
2	512	0
2	256	0
2	128	0
2	64	0
2	32	0
2	16	0
2	8	0
2	4	0
2	2	0
2	1	0



$$\begin{array}{r} 0.1024 \\ \times 8 \\ \hline 0.8192 \end{array}$$

$$\begin{array}{r} 0.8192 \\ \times 8 \\ \hline 6.5536 \end{array}$$

1:- Convert $(128.25)_{10}$ to Octal system

2:- Convert $(555.345)_{10}$ to Octal system

3:- Convert $(176.1024)_{10}$ to Octal system

$$\begin{array}{l} 0.1024 \times 8 = 0.8192 \rightarrow 0 \\ 0.8192 \times 8 = 6.5536 \Rightarrow 6 \\ 0.5536 \times 8 = 4.4288 \Rightarrow 4 \\ 0.4288 \times 8 = 3.4304 \Rightarrow 3 \\ 0.4304 \times 8 = 3.4432 \Rightarrow 3 \\ 0.4432 \times 8 = 3.5456 \Rightarrow 3 \\ 0.5456 \times 8 = 4.3648 \Rightarrow 4 \\ 0.3648 \times 8 = 2.9184 \Rightarrow 2 \\ 0.9184 \times 8 = 7.3472 \Rightarrow 7 \\ 0.3472 \times 8 = 2.7776 \Rightarrow 2 \\ 0.7776 \times 8 = 6.2208 \Rightarrow 6 \\ 0.2208 \times 8 = 1.7664 \Rightarrow 1 \\ 0.7664 \times 8 = 6.1312 \Rightarrow 6 \\ 0.1312 \times 8 = 1.0496 \Rightarrow 1 \\ 0.0496 \times 8 = 0.3968 \Rightarrow 0 \end{array}$$

$$\begin{array}{l} 0.25 \times 8 = 2.0000 \Rightarrow 2 \\ 0.00 \times 8 = 0 \end{array}$$

$$\begin{array}{r} 8 \overline{) 128} \\ 8 \overline{) 16} \\ \hline 2 \end{array}$$

$$(200.20)_8$$



1:- Convert $(786.25)_{10}$ to HexaDecimal system

2:- Convert $(4500.345)_{10}$ to HexaDecimal system

3:- Convert $(1982.1024)_{10}$ to HexaDecimal system

$$\begin{array}{r|l} 16 & 786 \\ \hline 16 & 49 \\ \hline & 3 \end{array} \begin{array}{l} \rightarrow 2 \\ \rightarrow 1 \\ \rightarrow 0 \end{array}$$

$(312.40)_{16}$

$$\begin{array}{l} 0.25 \times 16 = 4.0 \\ 0 \times 16 \Rightarrow 0 \end{array} \quad \begin{array}{l} 1 \times 16 = 16.0 \\ \underline{4} \end{array}$$



32100(0.03

Binary
Octal
Hexadecimal
Pattern Same
Decimal

1:- Convert $(101101.11001)_2$ to Decimal system

2:- Convert $(10101010.111)_2$ to Decimal system

3:- Convert $(1101110.1001)_2$ to Decimal system

$$1 \times \frac{1}{2^5}$$

32 + 8 + 4

$$\begin{array}{r} 101101.11001 \\ \hline 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \quad 2^{-3} \quad 2^{-4} \quad 2^{-5} \\ \hline (45.78)_{10} \end{array}$$

$$\begin{array}{r} 0.3 \\ 0.25 \\ 0.03 \\ \hline 0.78 \end{array}$$

$$\begin{array}{r} 0.24 \\ 0.12 \\ 0.06 \\ 0.03 \\ \hline 0.45 \end{array}$$

$$\begin{array}{r} 0.78 \times 8 = 6.24 \\ 0.24 \times 8 = 1.92 \\ \hline 8.16 \end{array}$$

61



$$(\text{Integer})^0 = 1$$

$$8^0 = 1$$

$$99^0 = 1$$

1:- Convert $(771.62)_8$ to Decimal system

$$8^0 = 1$$

2:- Convert $(1994.125)_8$ to Decimal system

$$(100000)^0 = 1$$

3:- Convert $(9090.9090)_8$ to Decimal system

$$\begin{matrix} 7 & 7 & 1 & \cdot & 6 & 2 \\ 8^2 & 8^1 & 8^0 & & 8^{-1} & 8^{-2} \end{matrix}$$

$$\underline{7 \times 8^2 + 7 \times 8^1 + 1 \times 8^0 + 6 \times 8^{-1} + 2 \times 8^{-2}}$$

$$\begin{array}{r} 2 \\ 64 \\ \times 7 \\ \hline 448 \end{array}$$

$$\begin{array}{r} 11.0 \\ 440 \\ 57 \\ \hline 505 \end{array}$$

$$\begin{array}{r} 1 \\ 456 \\ 55 \\ \hline 505 \end{array}$$

$$\begin{array}{l} 448 + 56 + 1 + 0.75 + 0.125 \\ \hline 505.875 \end{array}_{10}$$



$$\begin{array}{r} 1792 \\ 254 \\ \hline 2046 \end{array}$$

$$\begin{array}{ccccccc} 16^2 & 16^1 & 16^0 & 16^{-1} & 16^{-2} \\ 7 & F & E & . & 6 & 2 \\ \downarrow & \downarrow & \downarrow & & & \\ 7 & 15 & 14 & & & \end{array}$$

1:- Convert $(7FE.62)_{16}$ to Decimal system

2:- Convert $(FEFE.FFF)_{16}$ to Decimal system /

3:- Convert $(12BC.9090)_{16}$ to Decimal system

$$7 \times 16^2 + 15 \times 16^1 + 14 \times 16^0 + 6 \times 16^{-1} + 2 \times 16^{-2}$$

$$= 1792 + 240 + 14 + 0.377 + 0.007$$

$$\Rightarrow 2046.387$$

$$16^3 \cdot 8) 30 (0.37256 \quad 256 \times 1 \quad 256$$

$$\begin{array}{r} 16 \\ 46 \\ \hline 96 \\ 156 \\ \hline 256 \end{array}$$

$$\begin{array}{r} 15 \\ 128 \\ \hline 896 \end{array}$$

$$\begin{array}{r} 25 \\ 128 \\ \hline 1024 \end{array}$$

$$128) 1000 (0.0078125$$



Using Grouping

Since, there are only 8 digits (from 0 to 7) in octal number system, so we can represent any digit of octal number system using only 3 bit as following below.

Octal Digit Value	Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111



- Take binary number
- Divide the binary digits into groups of three (starting from right) for integer part and start from left for fraction part.
- Convert each group of three binary digits to one octal digit.

This is simple algorithm where you have to grouped binary number and replace their equivalent octal digit.



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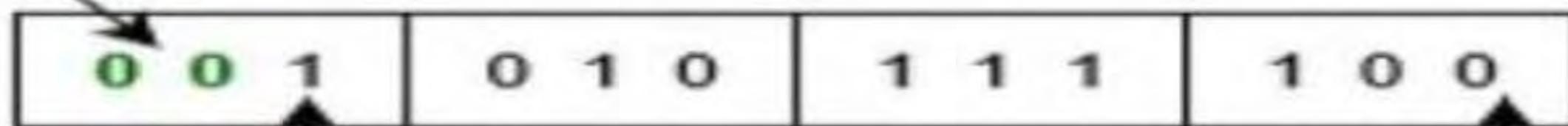
Topic : Binary to Octal Number



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Example-1 – Convert binary number 1010111100 into octal number. Since there is no binary point here and no fractional part. So,

These two 0's are added into MSB to complete group of 3 bits



9th position
(MSB)

0th position
(LSB)

Therefore, Binary to octal is.

$$\begin{aligned} &= (1010111100)_2 \\ &= (001\ 010\ 111\ 100)_2 \\ &= (1\ 2\ 7\ 4)_8 \\ &= (1274)_8 \end{aligned}$$



Topic : Binary to Octal Number



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Example-2 Convert binary number 0110 011.1011 into octal number. Since there is binary point here and fractional part. So,

Single 0 at MSB
can be ignored

Binary
point

These two 0's are added
into LSB to complete group of 3 bits



Therefore, Binary to octal is.

$$\begin{aligned}
 &= (0110\ 011.1011)_2 \\
 &= (0\ 110\ 011\ .\ 101\ 1)_2 \\
 &= (110\ 011\ .\ 101\ 100)_2 \\
 &= (6\ 3\ .\ 5\ 4)_8 \\
 &= (63.54)_8
 \end{aligned}$$

Left to Right
Binary to Octal

Right to Left
Pair

Extra

(-4)th position (LSB)

Extra

Binary to Octal

63.54

6 3 5 4



Using Grouping

Since, there are only 16 digits (from 0 to 7 and A to F) in hexadecimal number system, so we can represent any digit of hexadecimal number system using only 4 bit as following below.

Hexa	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111
Hexa	8	9	A	B	C	D	E	F
Binary	1000	1001	1010	1011	1100	1101	1110	1111

So, if you make each group of 4 bit of binary input number, then replace each group of binary number from its equivalent hexadecimal digits. That will be hexadecimal number of given number. Note that you can add any number of 0's in leftmost bit (or in most significant bit) for integer part and add any number of 0's in rightmost bit (or in least significant bit) for fraction part for completing the group of 4 bit, this does not change value of input binary number.

So, these are following steps to convert a binary number into hexadecimal number.



So, these are following steps to convert a binary number into hexadecimal number.

- Take binary number
- Divide the binary digits into groups of four (starting from right) for integer part and start from left for fraction part.
- Convert each group of four binary digits to one hexadecimal digit.

This is simple algorithm where you have to grouped binary number and replace their equivalent hexadecimal digit.



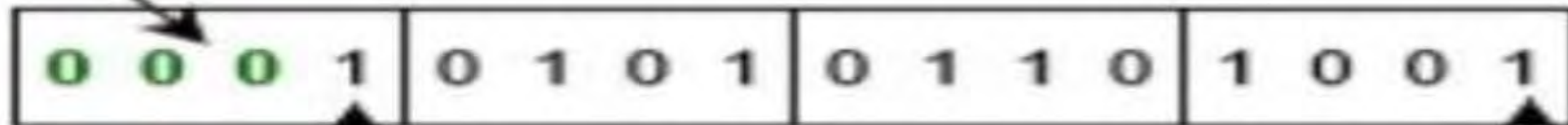
Topic : Binary to HexaDecimal Number



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Example-1 – Convert binary number 1010101101001 into hexadecimal number. Since there is no binary point here and no fractional part. So,

These three 0's are added into MSB to complete group of 4 bits



12th position
(MSB)

0th position
(LSB)

$$= (1010101101001)_2$$

$$= (1\ 0101\ 0110\ 1001)_2$$

$$= (0001\ 0101\ 0110\ 1001)_2$$

$$= (1\ 5\ 6\ 9)_{16}$$

$$= (1569)_{16}$$



Topic : Binary to HexaDecimal Number



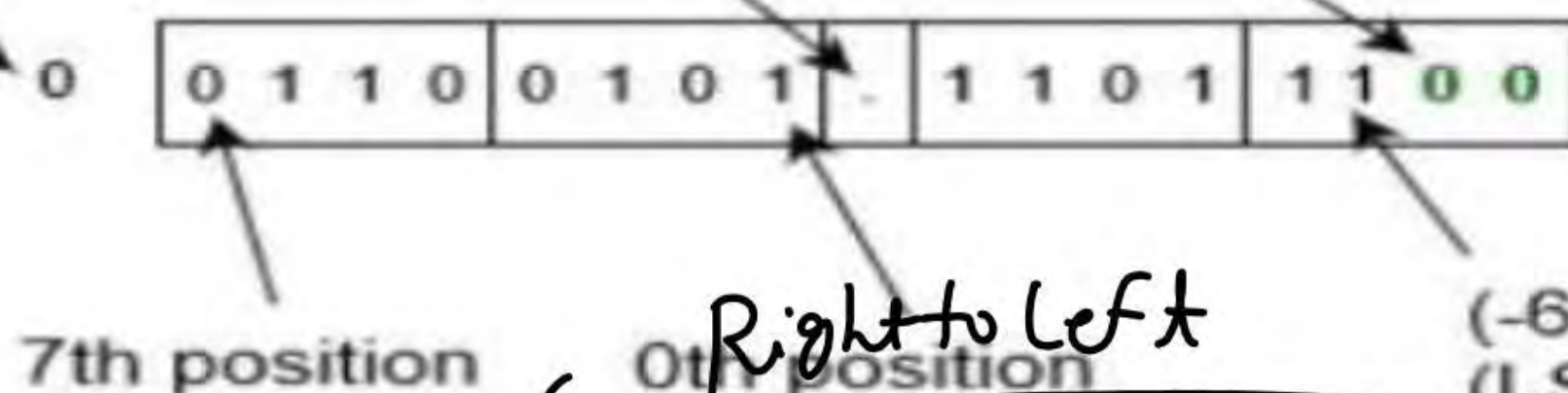
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Example-2 – Convert binary number 001100101.110111 into hexadecimal number. Since there is binary point here and fractional part. So,

Extra 0 at MSB
can be ignored

Binary
point

These two 0's are added
into LSB to complete group of 4 bits



Right to Left

$$\begin{array}{ccccccc} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1} & 2^{-2} \end{array}$$

26.25

Therefore, Binary to hexadecimal is,

$$\begin{array}{ccccccc} 00 & 00 & 01 & 10 & 10 & 11 & 01 & 11 & 01 & 11 \\ \hline & & 0 & & & & & & & \end{array}$$

(65.DC)₁₆

$$\begin{aligned} &= (001100101.110111)_2 \\ &= (0\ 0110\ 0101\ .\ 1101\ 1100)_2 \\ &= (0110\ 0101\ .\ 1101\ 1100)_2 \\ &= (6\ 5\ .\ D\ C)_{16} \\ &= (65.DC)_{16} \end{aligned}$$



What is Binary Addition

The binary addition operation works similarly to the base 10 decimal system, except that it is a base 2 system. The binary system consists of only two digits, 1 and 0. Most of the functionalities of the computer system use the binary number system. The binary code uses the digits 1's and 0's to make certain processes turn off or on. The process of the addition operation is very familiar to the decimal system by adjusting to the base 2.

Before attempting the binary addition process, we should have complete knowledge of how the place works in the binary number system. Because most of the modern digital computers and electronic circuits perform the **binary operation** by representing each bit as a voltage signal. The bit 0 represents the "OFF" state, and the bit 1 represents the "ON" state.

Rules of Binary Addition

Binary addition is much easier than the decimal addition when you remember the following tricks or rules. Using these rules, any binary number can be easily added. The four rules of binary addition are:

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 10$



$$\begin{array}{r} 1:- \quad 101010111 \\ + 101000011 \\ \hline \end{array}$$

$$\begin{array}{r} 2:- \quad 10000111010 \\ + 1010101010 \\ \hline \end{array}$$

$$\begin{array}{r} 3:- \quad 10101010 \\ + 10101010 \\ \hline \end{array}$$



Binary Subtraction is a mathematical operation and is one of the four fundamental operations of binary numbers. Binary subtraction is similar to decimal subtraction but the rules used are somewhat different.

- Binary subtraction is a fundamental idea in [binary operations](#).
- There are two components in binary subtraction: **0** and **1**.



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- Binary subtraction is a fundamental idea in [binary operations](#).
- There are two components in binary subtraction: **0** and **1**.

x	y	$x - y$
0	0	0
0	1	1 (Borrow 1)
1	0	1
1	1	0



Method 1: Subtraction of Binary Numbers without Borrowing

Step 1:-In binary, the number 8 is represented as $(1000)_2$ and the number 25 is represented as $(11001)_2$. Now subtract $(1000)_2$ from $(11001)_2$.

$$\begin{array}{r} 11001 \\ - 1000 \\ \hline \end{array}$$

Step 2: To subtract the numbers, use binary subtraction principles.

Let us begin this subtraction by subtracting the integers on the right and working our way up to the next **higher-order** digit.

To begin, subtract $(1-0)$. This equals 1.

Likewise, we proceed to the next higher-order digit and subtract $(0-0)$, which is **equal** to 0.

Again, subtract $(0-0)$, which is equals to 0. Then, we subtract $(1-1)$, the outcome is 0

As a result, the difference is 10001_2 .

$$\begin{array}{r} 11001 \\ - 1000 \\ \hline 10001 \end{array}$$



Method 1: Subtraction of Binary Numbers with Borrowing

Step 1:- In binary, the number 12 is represented as $(1100)_2$ and the number 26 is represented as $(11010)_2$. Now subtract $(1100)_2$ from $(11010)_2$. Put the numbers in the order given below

$$\begin{array}{r} 11010 \\ - 1100 \\ \hline \end{array}$$

Step 2: To subtract the numbers, use binary subtraction principles.

Start the subtraction from the rightmost digit and move toward the higher-order digits.

Subtract $(0 - 0) \rightarrow$ Result: 0

Next, subtract $(1 - 0) \rightarrow$ Result: 1

Then, subtract $(0 - 1)$:

Borrow 1 from the next higher order digit.

Result of $(0 - 1)$ after borrowing $\rightarrow 1$

Next, subtract $(1 - 1) \rightarrow$ Result: 0

Finally, subtract 0 (from the previous result) - 1 (borrowed) $\rightarrow$ Result: 1

Final binary difference: $(1110)_2$

$$\begin{array}{r} 11010 \\ - 1100 \\ \hline 1110 \end{array}$$



Binary Subtraction Using 1's Complement

- The number 0 represents the positive sign
- The number 1 represents the negative sign

Procedures for Binary Subtraction by 1's Complement

- Write the 1's complement of the subtrahend
- Then add the 1's complement subtrahend with the minuend
- If the result has a carryover, then add that carry over in the least significant bit
- If there is no carryover, then take the 1's complement of the resultant, and it is negative.



Topic : BINARY SUBTRACTION using 1'S Complement



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$$53 - 37$$

Ex-1 :- $(110101)_2 - (100101)_2$

subtrahend

=> Take 1's Complement of Subtrahend

(011010)

=> Add this 1's complement to with Minuend means

$$\begin{array}{r} 110101 \\ + 011010 \\ \hline \end{array}$$

1 001111 if carry 1 comes Discard it and carry 1 to add

Remaining output.

$$\begin{array}{r} 001111 \\ + 1 \\ \hline \underline{010000} \text{ result} \end{array}$$



Ex-1 :- $(101011)_2 - (111001)_2$

subtrahend

=> Take 1's Complement of Subtrahend

(000110)

=> Add this 1's complement to with Minuend means

$$\begin{array}{r} 101011 \\ + 000110 \\ \hline \end{array}$$

$11000\underline{1}$ if carry 1 not comes then takes 1's complement again

$00\underline{1110}$ output result



Binary Subtraction Using 2's Complement

Binary subtraction with 2's complement entails adding the complement of the subtrahend (the number being subtracted) to the minuend (the amount being removed). Here's an illustration:

Let's subtract $(100010)_2$ from $(110110)_2$. In this case, the binary equivalent of 34 is $(100010)_2$ whereas the binary equivalent of 54 is $(110110)_2$.

Step 1: First identify the Minuend and Subtrahend. In this, this Minuend is $(110110)_2$ and subtrahend is $(100010)_2$.

Step 2: Find the 1's complement of the subtrahend. The subtrahend is $(100010)_2$ and after 1's complement it is $(011101)_2$. Now add 1 to the 1's complement $(011101)_2 + 1 = (011110)_2$. So the 2's Complement is $(011110)_2$.

Step 3: Now add the minuend and the 2's complement of the subtrahend.



110110 (Minuend)

+ 011110 2's Complement of Subtrahend

1010100

THANK YOU